

A deep learning-geospatial analytics fusion framework for predicting land use land cover dynamics in eastern region of India

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ABSTRACT

Globally, rapid urbanisation and unregulated urban expansion have driven widespread land use and land cover (LULC) changes, resulting in deforestation, loss of vegetation, and depletion of water bodies. In India, cities and peri-urban areas are experiencing similar transformations, making it essential to monitor and manage urban growth effectively. This study aims to analyse LULC changes in Berhampore, West Bengal, from 2001 to 2023, identify key drivers of urban expansion, and predict future land use for 2033 using remote sensing, GIS-based spatial analysis, and deep learning techniques (ANN and CA models). The results show a sharp increase in built-up areas from 20.39 sq. km (2001) to 43.85 sq. km (2023), leading to significant reductions in vegetation (-52.13 sq. km) and water bodies (-1.74 sq. km). Forecast for 2033 shows continued urban development where agricultural territories represent sixty-two percent of the land while forests face a fifty percent reduction thus boosting risks of biodiversity depletion along with enhanced flood potential. This research proves that land sustainability policies and green infrastructure planning combined with strategic urban strategies must urgently develop to balance development with environmental endurance. This research shows that ANN and CA operate as predictive methods to enhance LULC accuracy by helping administrators execute measurement policies. This study incorporates deep learning architecture parts that integrate CNNs and LSTMs to establish advanced LULC modelling frameworks which excel at spatial-temporal pattern detection. Future research on urban growth prediction will benefit from the proposed structures built to improve predictive models by integrating hierarchical features based on long-term dependency according to academic publications.

1. Introduction

The patterns of land use and land cover (LULC) have been modified by human activities for thousands of years, but in the last century, those changes have increased significantly, as the population has grown at a fast rate, food needs have risen, and socio-economic development has been developed [1,2]. The changes cause severe ecosystem changes, which influence hydrological processes, biophysical interactions, and environmental stability in complex socio-ecological structures [3,4]. Conversion of natural landscapes into croplands and urban settlements has increased the stressors faced by food security and the sustainability

of the environment with almost 62% of the land areas being transformed [5,6]. Increase in population, growth in urban areas, changes in farming, energy demand, deforestation and massive land conversion processes are the main causes of LULC change [7,8]. The insights into these changes demand elaborate modelling structures that are able to explain the interconnected dynamisms of social, environmental, institutional and economic forces [9].

Computational approaches for explaining and predicting LULC change are commonly categorised based on their spatial and temporal characteristics, as well as inductive-deductive reasoning and pattern-based or agent-based modelling strategies [10]. In these models,

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multi-source data are becoming more integrated, which includes remotely sensed data and socio-economic data in geographic information system (GIS) analytics to determine the spatio-temporal land use change [11,12].

In India, there is a challenge of LULC as a result of rapid urbanisation, and as such, policies have to be brought into action in time with the aid of sound spatial planning strategies [13]. India is already the leading agricultural country, with a population of more than 1.44 billion, and a scarce per-capita land supply per person, which is putting pressure on both natural and agricultural scenery. Therefore, LULC changes need systematic tracking and projections to know their drivers, effects and trends into the future in accordance with sustainable development goals and sustainable development goals (SDGs) [14].

The recent development in remote sensing (RS) and GIS has significantly contributed to the capacity to track dynamics of LULC by providing effective spatial and temporal measurements of urban growth [15–17]. These technologies offer affordable means of examining urban growth patterns and their effects to the environment, and through which it facilitates decision-making at both local, regional and national levels [18–20]. Sampling satellite data like Landsat, QuickBird, and Google Earth has also reinforced the mapping of LULC and detection of changes [21–23]. Remotely sensed information has now become essential in determining the land transformation and climate variability in the data-scarce developing regions [24–26]. The analytical frameworks based on GIS are effective in consolidating various datasets and enable thorough assessment of the spatial and temporal LULC changes [27–29].

However, more recently, artificial intelligence (AI) and machine learning (ML) methods have been integrated in RS and GIS-based LULC analysis, enhancing the level of automation of the classification, pattern detection, and predictive accuracy [30,31]. Predictive modelling is significant in LULC studies as it helps to gain an overview of future urban growth, agricultural transition, and forest dynamics [32,33]. Markov Chain and Cellular Automata (CA) models are some of the most popular models that have been effective in simulating spatial dynamics and future land use conditions [34,35]. CA-based models capture neighbourhood effects and transition rules, enabling the simulated changes in LULC to be more realistic in space [36,37]. Recently, CA performance has been enhanced again with the use of suitability maps as an alternative to conventional transition probability matrices based on spatial variables [38,39]. These applications have facilitated the use of RS data and GIS-based spatial analysis in data-based decision-making processes in terms of urban planning, agricultural management, and environmental governance [40,41]. Artificial neural networks (ANN), support vector machines (SVM), random forest (RF), and Decision Trees are ML-based classifiers that have shown good performance in classification and prediction tasks of LULC and tend to achieve high precision even when learning heterogeneous data [42–44], as well as when learning disparate data [45,46]. Nevertheless, single models are characterised by certain limitations, which have given rise to an increased interest in hybrid modelling methods that result in the strengths that are complementary to each other. The implementation of hybrid models like MLP-CA-Markov and CA-ANN has demonstrated to lead to improved predictive performance and spatialization of urban expansion and land use conversion [47–49].

Recent development in the field of deep learning provides new possibilities to address the existing challenges of LULC modelling. Convolutional neural networks (CNNs) are specifically good at finding hierarchical spatial features in satellite images, whereas long short-term memory (LSTM) networks are better at addressing temporal relationships in sequential land covers [50]. Although proven to be effective, the use of deep learning architectures in extensive geospatial simulation models is somewhat scarce in any applied LULC literature, particularly with medium-sized cities in eastern India.

The proposed research will fill this gap by combining remote sensing, GIS-based spatial analysis, and predictive modelling to examine the historical LULC changes (2001–2023) and forecast the future land use

pattern in 2033 in Berhampore, West Bengal. The paper uses an ANN-based CA model to conduct a spatial simulation and introduces a hybrid conceptual architecture of deep learning to improve the modelling of spatio-temporal dynamics of LULC using a CNN and LSTM combination. This study will assist in evidence-based urban planning and sustainable land-use policy development in fast urbanising areas by combining traditional geospatial methods with modern learning-based methods [51].

The primary aim of this study is to develop and apply an integrated geospatial-machine learning framework to (i) quantify long-term LULC transformations in a rapidly urbanising medium-sized city of eastern India (Berhampore, West Bengal), and (ii) simulate future land use dynamics using an ANN-Cellular Automata modelling approach. Beyond conventional LULC prediction, the study further aims to conceptually advance existing methodologies by proposing a hybrid deep learning architecture that combines CNNs and LSTM networks to better capture spatial heterogeneity and temporal dependency in urban land transformation processes. This integrated approach addresses a critical gap in applied LULC modelling studies, which remain limited for medium-sized Indian cities despite their accelerating urban growth and planning relevance.

2. Materials and methods

The research is an amalgamation of multi-temporal remote sensing methods, spatial analysis of GIS, and predictive modelling to evaluate the historical LULC evolution between 2001 and 2023 and simulate a future land use scenario by 2033.

2.1. Study area

Berhampore is a municipal city in the Murshidabad district of West Bengal, India, and serves as the district administrative headquarters (Fig. 1). It is the seventh-largest urban agglomeration in the state, geographically located between 23°50'N–24°10'N latitude and 88°10'E–88°25'E longitude. The city lies along the eastern bank of the Bhagirathi River, a major tributary of the Ganges (Hooghly River downstream), approximately 200 km northeast of Kolkata. The region experiences a tropical climate, with temperature extremes ranging from 8.8°C in winter to 42.4°C during summer.

Administratively, Berhampore comprises 25 municipal wards. The population increased from 195,223 in 2011 to approximately 253,000 in 2021 and is projected to reach nearly 334,000 by 2031 (Fig. 1e). Rapid demographic growth, infrastructural expansion, and administrative importance have resulted in pronounced LULC transformations, making the city a suitable case for urban growth modelling.

2.2. Data sources and materials

The present research combines the multi-temporal remote sensing, GIS-based spatial variables, and predictive modelling in analysing the LULC change between 2001 and 2023 and predicting future scenarios by 2033. The satellite-based LULC maps were created based on the Landsat-5 TM and Landsat-8 OLI/TIRS satellite imagery over the two years. The use of ancillary spatial data, such as population density, road networks, rivers, railways, and topographic slope based on the Shuttle Radar Topography Mission (SRTM) DEM, was used to facilitate modelling and simulation processes.

The maximum likelihood algorithm was used to do supervised classification in ArcGIS 10.8. LULC change detection analysis was then performed in order to measure changes across classes over time. All datasets used will be summarised in Table 1. The general project methodology process followed in this study is presented in Fig. 2, and the suggested hybrid-based deep learning prediction algorithm is outlined in Table 2.

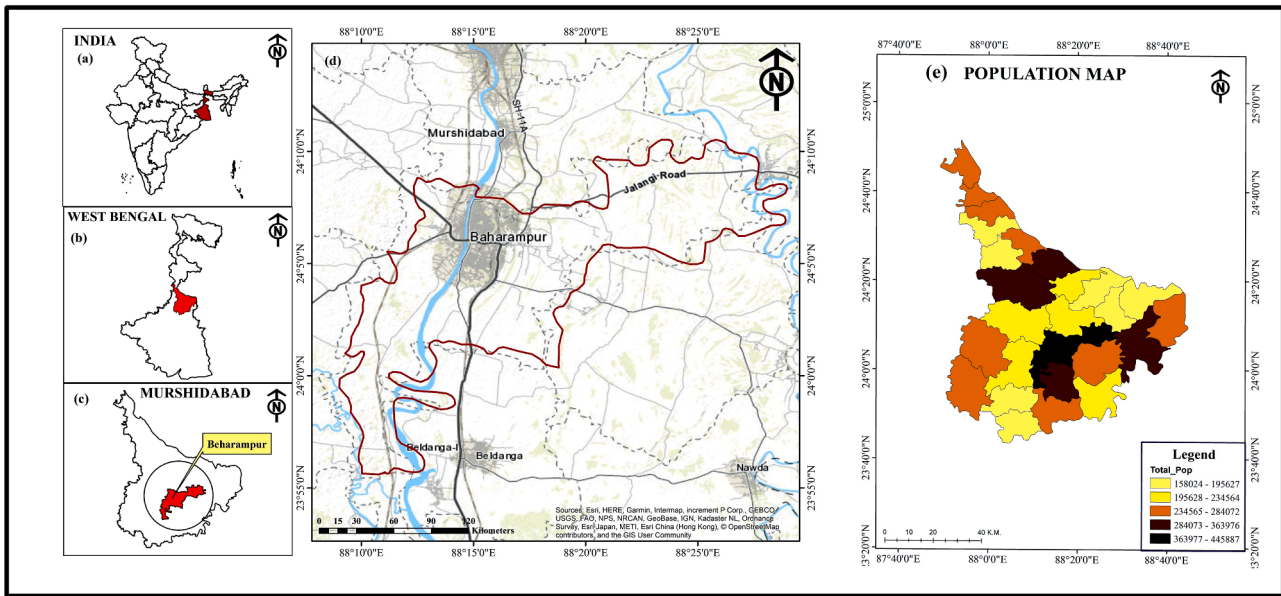


Fig. 1. (a) India map, (b) West Bengal, (c) Murshidabad, (d) Study area map, (e) Population map of Murshidabad.

Table 1

Data Sources for Thematic Layers.

Attribute	Source
Population	Census of India
Road, River, Railway	Google Earth Pro
Slope	SRTM DEM Data, USGS
Land use land cover	Landsat-5, Landsat-8

2.3. LULC classification and change detection

The multi-temporal satellite images were first pre-processed and classified into five major LULC classes, namely built-up area, agricultural land, vegetation, water bodies, and fallow land. Visual interpretation and ancillary reference data were used to select the training samples. The accuracy of classification was determined using the conventional accuracy evaluation algorithms before the analysis of change detection. The LULC transitions between 2001 and 2023 were quantified by using post-classification comparison. Statistical data of each of the classes was produced to determine the spatial growth or decrease, which formed the input in predictive modelling.

2.4. Preparation of spatial driving variables

The major spatial drivers that affected the population density of urban expansion that included slope, distance to roads, distance to rivers, and distance to railways, were ready within a GIS setting. The conversion of all the vector layers into raster form and resampling them to a consistent spatial resolution of 30 m, as well as projecting them to the WGS 1984 UTM Zone 45N coordinate system, was done to make them spatially compatible [52]. All the variables were normalised to ensure that they have the same level of contribution to the modelling process and also to prevent any bias due to the scale in estimating the transition potential.

2.5. Land use modelling using MOLUSCE

The modules for land use change evaluation (MOLUSCE), plugin in QGIS was used to do LULC simulation [53]. MOLUSCE uses a combination of historical LULC maps and spatial driving factors to estimate potential of transition and predict future land use patterns. ANN were

applied to transition potential modelling because these networks are capable of learning non-linear responses to observed LULC transitions in relation to spatial variables. To use the neighbourhood effects and spatial contiguity, the ANN-generated transition potential maps were incorporated into a CA simulation. To create probabilistic future LULC scenarios, CA simulation was applied via Monte Carlo simulation.

The modelling framework took into consideration population dynamics such as census data in the past and future population projections through the incremental increase method to derive population dynamics until the year 2031 [54]. These spatial driving variables (slope, distance to roads, rivers, railways and population density) were input factors in the ANN-based transition potential modelling within the MOLUSCE framework, and thus had a direct influence on land use transitions during Cellular Automata simulation. The general modelling process is depicted in Fig. 2.

2.6. Model validation

The accuracy measures based on kappa were also used to assess the performance of the models, such as overall kappa, kappa location and kappa histogram. The analysis of misses and false alarms was done to determine spatial agreement between observed and simulated LULC maps. These validation tests guaranteed the soundness of transition regulations and simulation results [55,56].

2.7. Proposed deep learning-based hybrid framework

To further improve the processing of spatio-temporal dynamics of LULC, the current work suggests a hybrid prediction system that will be conducted by deep learning (Table 2). The framework is also in the form of a methodological extension that is to be used in conducting further implementation in the future and is set to complement the ANN-CA modelling approach that is being used in the study [57]. The proposed framework adheres to an end-to-end workflow in sequence; it combines the learning of spatial features, temporal sequence modelling, traditional ANN-CA prediction, ensemble fusion, and uncertainty estimation. The algorithm, as summarised (Table 2) starts with the multi-temporal satellite images and spatial variables pre-processing phase, wherein all of the inputs are normalised, rasterised, and the standardisation to a uniform spatial resolution and spatial coordinate system. The processed satellite data are subsequently arranged into time series to clearly

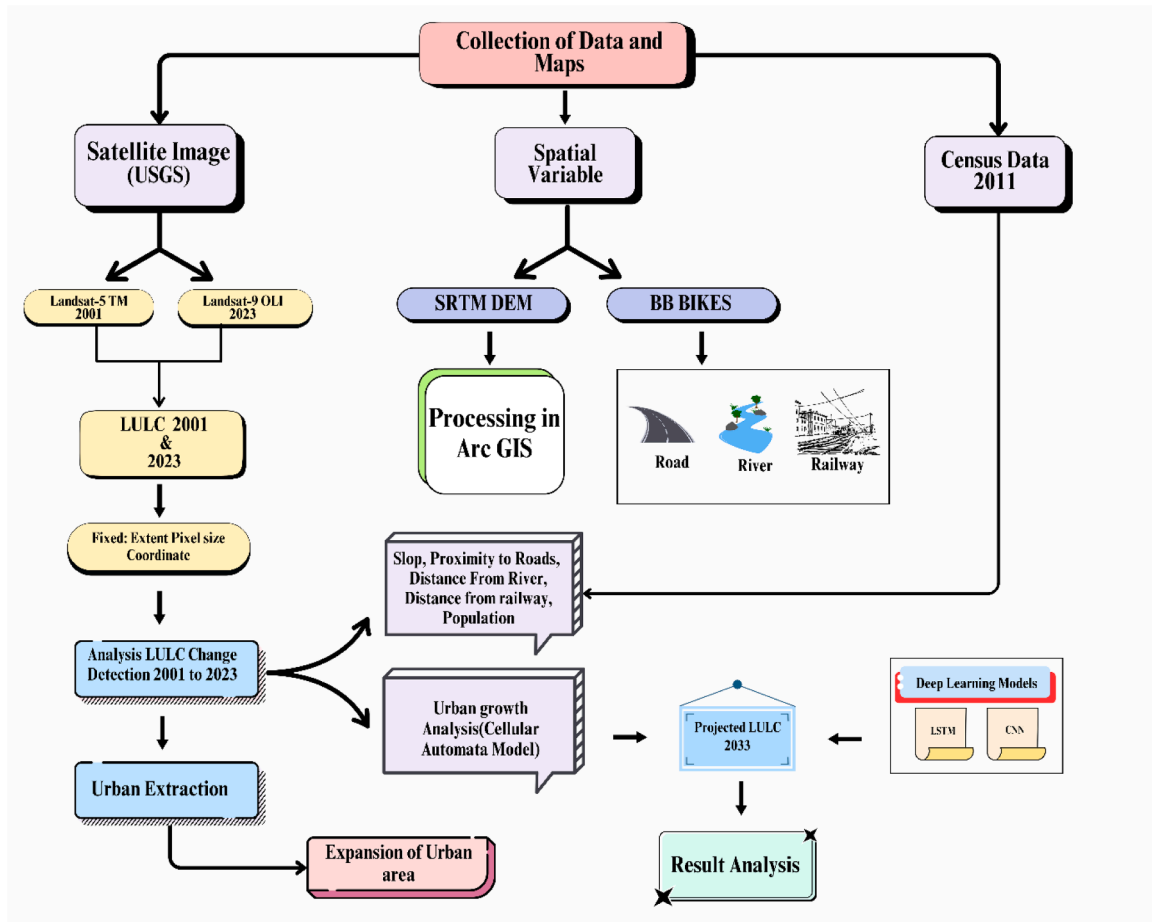


Fig. 2. Methodology Workflow.

demonstrate how the land cover changes with time.

At the second step, the temporal images are analysed separately through a CNN to obtain hierarchical spatial feature representations. The resulting spatial feature maps are then fed sequentially as inputs to a LSTM network, which estimates the temporal variations through the sequence of multi-year images and yields a deep learning-based LULC prediction (DL_pred). Simultaneously, a conventional prediction branch is run with the help of historical LULC maps and spatial driving variables, in which transition probabilities are obtained with the help of a Markov Chain and spatial suitability is estimated with the help of ANN-CA modelling. A weighted ensemble strategy is used to combine the results of the deep learning and traditional branches to generate the final LULC prediction. In order to determine predictive reliability, Monte Carlo Dropout is used in the inference stage of the CNN-LSTM network. Several stochastic forward passes are run to obtain a distribution of prediction, the mean prediction and the underlying uncertainty (standard deviation) are then calculated. The workflow proposed is such that not only are the spatially explicit LULC forecasts obtained, but the location of the prediction uncertainty is also quantified. The Table 2 provides the entire description of operations, inputs, and outputs.

2.7.1. CNN-based spatial feature extraction

CNNs are proposed for hierarchical spatial feature extraction from satellite imagery. Given a two-dimensional input image I and a convolution kernel K , the convolution operation at pixel location (i,j) is defined as:

$$(I * K)_{ij} = \sum_m \sum_n I_{i+m, j+n} \cdot K_{m,n}$$

This operation enables the automated detection of spatial features

such as edges, textures, and structural patterns that are critical for distinguishing LULC classes. Through successive convolution and ReLU activation layers, the CNN transforms raw satellite imagery into high-level spatial feature maps that encode spatial heterogeneity relevant to land cover classification [58,59]. These feature maps serve as the spatial inputs for subsequent temporal modelling.

2.7.2. LSTM-based temporal modelling

LSTM networks are proposed to model temporal dependencies in sequential LULC data [60,61]. The LSTM component processes the sequence of spatial feature maps extracted by the CNN to capture long-term temporal relationships in land cover transitions. The internal operations of an LSTM cell are defined as:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \cdot \tanh(C_t)$$

where σ denotes the sigmoid function, h_t is the hidden state, C_t is the cell state, and x_t represents the input at time step t . By updating its internal states through these gated mechanisms, the LSTM captures the temporal evolution of spatial patterns and produces a deep learning-based LULC

Table 2

Algorithm DL_Hybrid_LULCPrediction.

Algorithm1
Input: SatelliteImages $\{I_1, I_2, \dots, I_t\}$, SpatialVars, HistoricalLULC, $\alpha \in [0,1]$, T Output: LULC_pred, Uncertainty Begin // 1. Preprocess Data // Normalise radiometric values, rasterise inputs, and stack // multi-temporal images into temporal sequences For t = 1 to T : $I_t \leftarrow \text{Normalize}(\text{Standardize}(\text{Rasterize}(I_t)))$ // 2. CNN Feature Extraction // Extract spatial feature maps F_t from each image I_t // using convolution and ReLU layers For t = 1 to T : $F = \text{CNN}(I_t)$ where $\text{CNN}(I_t) = \sigma(W_k * I_t + b_k)$ for layers $k=1..K$ (convolution + ReLU) // 3. LSTM Temporal Modelling // Model temporal evolution of spatial features across time // and generate deep learning-based LULC prediction (DL_pred) $h_0 = 0, C_0 = 0$ For t = 1 to T: $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$ // Forget gate $i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$ // Input gate $\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$ // Candidate cell state $C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$ // Cell state update (* = Hadamard) $o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$ // Output gate $h_t = o_t * \tanh(C_t)$ // Hidden state $DL_{pred} = h_T$ // Final DL prediction // 4. Traditional Prediction // Generate CA-ANN based prediction using transition matrix // and spatial suitability weights P = Transition Matrix (Historical) // Markov chain S = Spatial Weights (Spatial Vars) //logistic regression $Trad_pred = P \times S$ // Matrix multiplication // 5. Ensemble Integration // Fuse DL and CA-ANN predictions using weighted averaging $LULC_pred = \alpha \cdot DL_pred + (1-\alpha) \cdot Trad_pred$ // 6. Uncertainty via Monte Carlo Dropout // Estimate spatial uncertainty using stochastic forward passes $MC_preds = []$ For mc = 1 to T: Enable dropout in CNN/LSTM $LULC_mc = \text{LSTM}(\text{CNN}(I_t, I_t))$ // Stochastic forward pass $MC_preds.append(LULC_mc)$ $\text{Uncertainty} = \sqrt{E[(MC_preds - E[MC_preds])^2]}$ // Std deviation $LULC_pred = E[MC_preds]$ // Mean prediction Return LULC_pred, Uncertainty End

prediction (DL_pred).

2.7.3. Ensemble integration and uncertainty quantification

The final LULC prediction is obtained through ensemble integration of the deep learning output and the traditional ANN-CA prediction. A weighted averaging scheme is applied as:

$$LULC_{pred} = \alpha \cdot DL_{pred} + (1 - \alpha) \cdot Trad_{pred}$$

where $\alpha \in [0, 1]$ controls the relative contribution of the deep learning and traditional modelling components [62,63]. This ensemble strategy leverages the complementary strengths of data-driven spatio-temporal learning and rule-based spatial simulation.

Prediction uncertainty is quantified using Monte Carlo Dropout, where dropout layers within the CNN-LSTM network are activated during inference to perform multiple stochastic forward passes. The resulting distribution of predictions is used to compute uncertainty as:

$$\text{Uncertainty} = \sqrt{E[(MC_{pred} - E[MC_{pred}])^2]}$$

This uncertainty measure provides spatial confidence information associated with future LULC predictions, supporting informed decision-making in urban planning and land management [64,65]. The complete

algorithmic workflow corresponding to this framework is summarised in the Table 2.

3. Results

3.1. Spatial driving variables and correlation analysis

Spatial driving variables such as road accessibility, railway accessibility, river accessibility and slope were ready to enable LULC transition modelling. The Fig. 3 shows the spatial distribution of these variables in the study area. The Pearson correlation was carried out to determine the relationship between these spatial variables and observed transitions in LULC. The values of correlation were all positive; this means that all four variables contributed to the change processes of the land use. The correlation study supports the applicability of the chosen variables to be included in the ANN-CA modelling framework.

3.2. Land use and land cover changes (2001-2023)

It was found that Berhampore and its surroundings experienced substantial changes in LULC during the period between 2001 and 2023. Table 3 summarises the area statistics of the classes and percentage

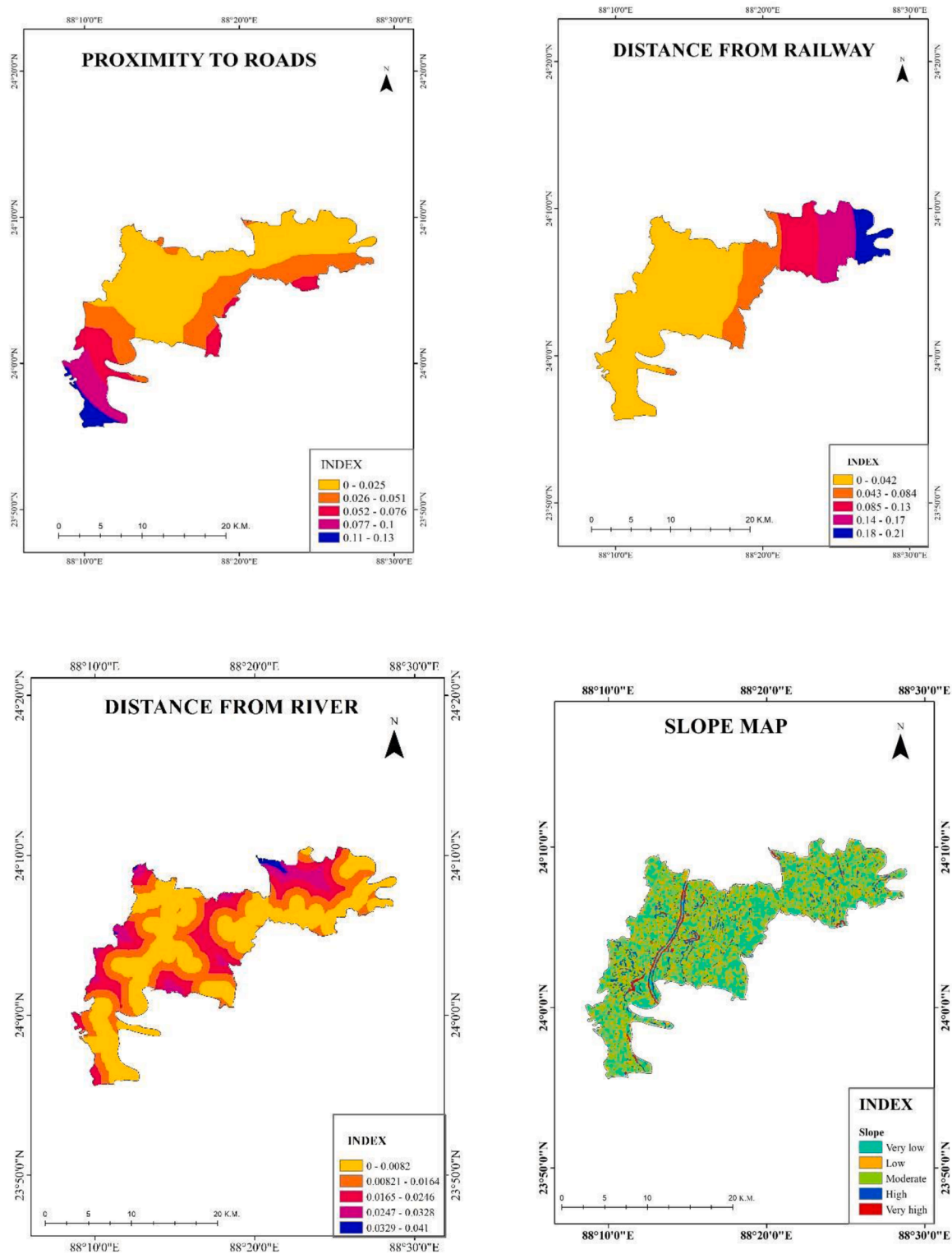


Fig. 3. Spatial variable maps (a) Proximity to roads; (b) Distance from railway; (c) Distance from river; (d) Slope.

change by class, whereas the spatial distribution of the LULC classes and transitions are represented in Figures 4-7. Farmland over the period 2001 to 2023 developed to 225.41 km² (69.54%), which is the net growth of 23.09 km² (7.12%). The proportion of built-up areas grew by 20.91 km² (6.45%) in 2001 to 43.85 km² (13.53%) in 2023, corresponding to an increase of 27.94 km² (7.08%). Vegetation cover declined substantially, decreasing from 91.43 km² (28.21%) to 39.30 km² (12.12%), a net loss of 52.13 km² (16.09%). The water bodies also

experienced a decrease of 1.74 km² (0.54%), and the fallow land experienced an increase of 5.84 km² (1.41%).

These changes can be represented by the spatial patterns as Fig. 4 demonstrates, and each of the LULC maps of 2001 and 2023 is provided separately in Figs. 5 and 6, respectively. The scale and pattern of transitions by the classes are also depicted in Fig. 7.

Table 3
Land use class area distribution and changes (2001-2023).

Class	2001 (sq. km)	2023 (sq. km)	Change (sq. km)	2001 (%)	2023 (%)	Change (%)
Water body	9.49	7.75	-1.74	2.93	2.39	-0.54
Built-up	20.91	43.85	27.94	6.45	13.53	7.08
Vegetation	91.43	39.3	-52.13	28.21	12.12	-16.09
Agricultural land	202.32	225.41	23.09	62.42	69.54	7.12
Fallow land	2.01	7.85	5.84	1.01	2.42	1.41

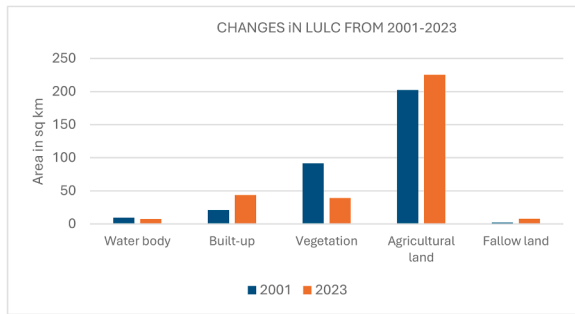


Fig. 4. Changes in LULC from 2001-2023.

3.3. Built-up area expansion

To measure the dynamics of urban growth, the built-up areas were isolated between the classified LULC maps of the years 2001 and 2023. The Fig. 8 shows the spatial distribution of the built-up land of both years and the expansion trend. The area of built-up land expanded to 43.85 km² in 2023, which was 113.5% higher in the 22 years compared to 20.39 km² in 2001. The growth took place mainly in major transportation routes and along the periphery areas around the urban centre.

3.4. Future LULC simulation for 2033

Simulation of future LULC patterns for 2033 using an integrated ANN-CA modelling on the basis of observed transitions and spatial driving variables was simulated. The forecasted LULC distribution is in Fig. 9. According to the simulation, built-up areas are going to continue to increase, especially around the current urban areas. The agricultural land is estimated to stabilise at around 62% of the total area. Forest cover is predicted to remain limited, accounting for nearly 5% of the study area. According to the simulated outcomes, there is also growing spatial continuity of urban land use, or progressive densification and outward expansion. The ANN-CA modelling framework was built in the MOLUSCE plugin, which uses QGIS, to produce the projected LULC map in 2033 using the historical LULC transitions (2001-2023) as well as the spatial driving factors. The hybrid approach, which is based on deep learning and offered in the present work of the study, is introduced as the conceptual methodological extrapolation to improve future spatio-temporal prediction without direct application in the present analysis to create the simulation of 2033.

4. Discussion

This research paper aimed to cope up with one of the major shortcomings of current LULC studies on the rapidly urbanising medium-sized Indian cities, that is, the absence of integrated modelling frameworks which can combine spatially explicit geospatial analysis with future urbanisation forecasting models that can reflect both historic land transitions and future urbanisation expansion. Using an ANN-CA modelling framework with support of multi-temporal remote sensing

and GIS data, the study offers a systematic explanation of the observed trends in LULC in Berhampore and assesses the future development of the land use.

The findings show that Berhampore has experienced a significant change in LULC in the last 20 years, and this has mainly been brought about by the high rate of urban growth. The area of built-up land has increased more than twice in the period between 2001 and 2023, and the natural vegetation has significantly reduced, as well as water bodies have decreased, due to the land conversion processes that are driven by urbanisation [66]. These trends are in line with the wider trends of urbanisation that have been recorded in various Indian cities with uncontrolled or semi-controlled development.

The use of vegetated and open land into impervious surfaces has significant effects on the environment. Past researches have attributed such urban growth trends to the change in surface energy balance, rise in land surface temperatures, and destabilisation of hydrological processes [67,68]. Despite this study not having any climatic feedback explicitly modelled, the spatial coverage and scale of the realised LULC change in Berhampore indicate an augmenting environmental pressure in line with these known results.

Methodologically, the hybrid approach of ANN-based transition potential modelling and Cellular Automata simulation was found to be useful not only in obtaining the non-linear relationships between spatial driving variables, but also in explaining spatial dynamics at a neighbourhood level. The ANN-CA framework is also able to represent the urban diffusion processes in a more realistic way, as it uses historical transition behaviour and spatial dependencies, unlike purely statistical or rule-based approaches. The modelling approach fills an established gap in traditional LULC research, in which either space interactions or non-linear drivers are typically poorly modelled.

More recent developments in deep learning also point to the potential of improved LULC prediction that can reflect complex spatial hierarchies and temporal associations [69,70]. In this regard, the conceptual hybrid framework suggested in this study, that combines CNN-based spatial feature extraction with LSTM-based temporal modelling, is a continuation of the current geospatial framework, as it aims at overcoming the limitations of the existing frameworks. Combined ensemble approaches with hybrid deep learning results and conventional models have been demonstrated to enhance predictive robustness and stability in complex land system forecasting [71,72]. Despite the conceptual nature of the deep learning framework described in this paper, it is introduced in direct relation to the research gaps in recent literature, where the lack of application of spatio-temporal deep learning architectures in practicable urban LULC research is observed. Such frameworks, which may be implemented in the future, especially through cloud-based systems like Google Earth Engine, may allow near real-time monitoring as well as adaptive urban planning [73,74].

Other than methodological issues, the functioning LULC modifications recorded in Berhampore are influenced by the interplay of environmental and socioeconomic drivers. The interaction between local forces, including seasonal floods, depletion of groundwater, and soil erosion, and population increase, housing needs, and population development determine land conversion patterns. These relations complicate the land management activities and increase exposure to environmental degradation. Similar research in emerging urban areas indicates similar patterns with agricultural and green lands increasingly changing into urban areas [75–77]. Rapid urbanisation in Indian cities like Bangalore and Hyderabad has led to massive destruction of the green cover and division of ecological spaces. Additional international case studies in Shanghai and Jakarta also show that economic growth and population movement also enhance LULC transitions and related environmental issues [78,79]. Despite these more general trends in urbanisation nationally and globally, the fact that Berhampore is still heavily reliant on the growth of agricultural lands and is subject to periodic flooding makes it increasingly important to discuss the location-specific land management techniques as opposed to the generic

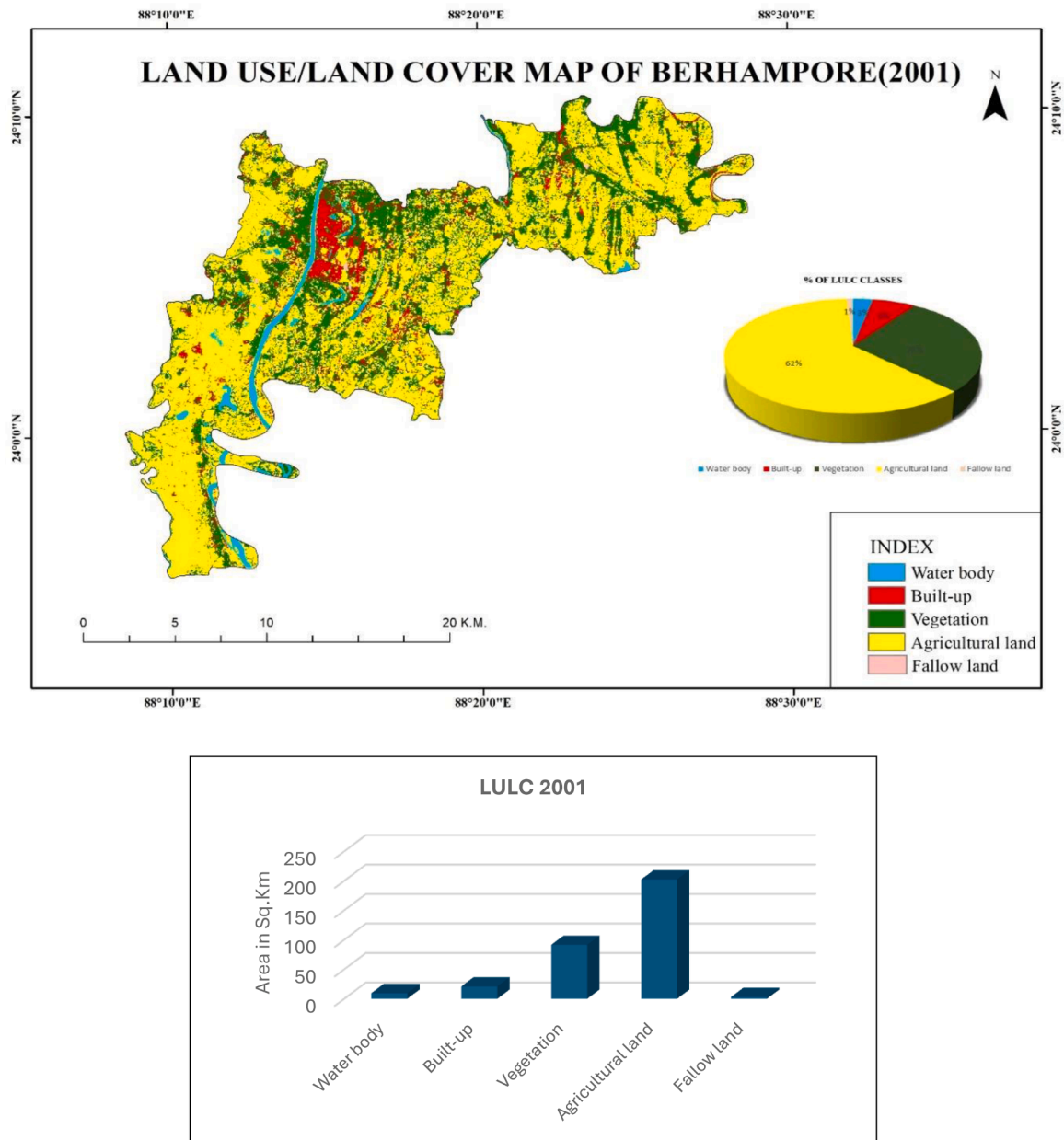


Fig. 5. LULC of the study area in 2001 (a) Map; (b) Area in sq. km.

urban planning solutions.

4.1. Implications, limitations, and future research directions

This study has demonstrated the relevance of the predictive LULC modelling to the city planning process to predict the future land use pressures. However, certain limitations must be acknowledged. The analysis is based on the historical satellite monitoring 2001-2023 and excludes the real-time monitoring potentials. Furthermore, specific socioeconomic forces like industrialisation, population trends and income dynamics were not explicitly incorporated because of the limitations in the data.

Future studies ought to incorporate socio-economic changes, climatic conditions and high-frequency satellites to come up with more comprehensive and dynamic models of urban growth. The data relevance and applicability of predictive LULC frameworks can also be enhanced by integrating participatory planning methods and community-based data. The ability to have active and resilient urban

land management would also be enhanced by expanding the proposed deep learning framework to fully operational implementations.

5. Conclusion and recommendations

This paper recorded significant LULC changes in Berhampore, West Bengal, over 2001 -2023 and showed that an ANN-CA modelling structure could be used to simulate future urban development. The findings indicate that the built-up areas are increasing rapidly at the expense of vegetation cover and the water bodies with a resultant strain on the natural landscapes. The projection LULC scenario of 2033 is an indication of further urbanisation, farming land as a ruling class, and low forest cover, which points to the possible environmental stresses in the long-term, as long as the present-day land use rates persist. The associated spatial patterns highlight the role of the (infrastructure) vicinity and population density on urban development processes. The combination of multi-temporal data with remote sensing, GIS-based spatial analysis, and ANN-CA modelling offers a powerful and

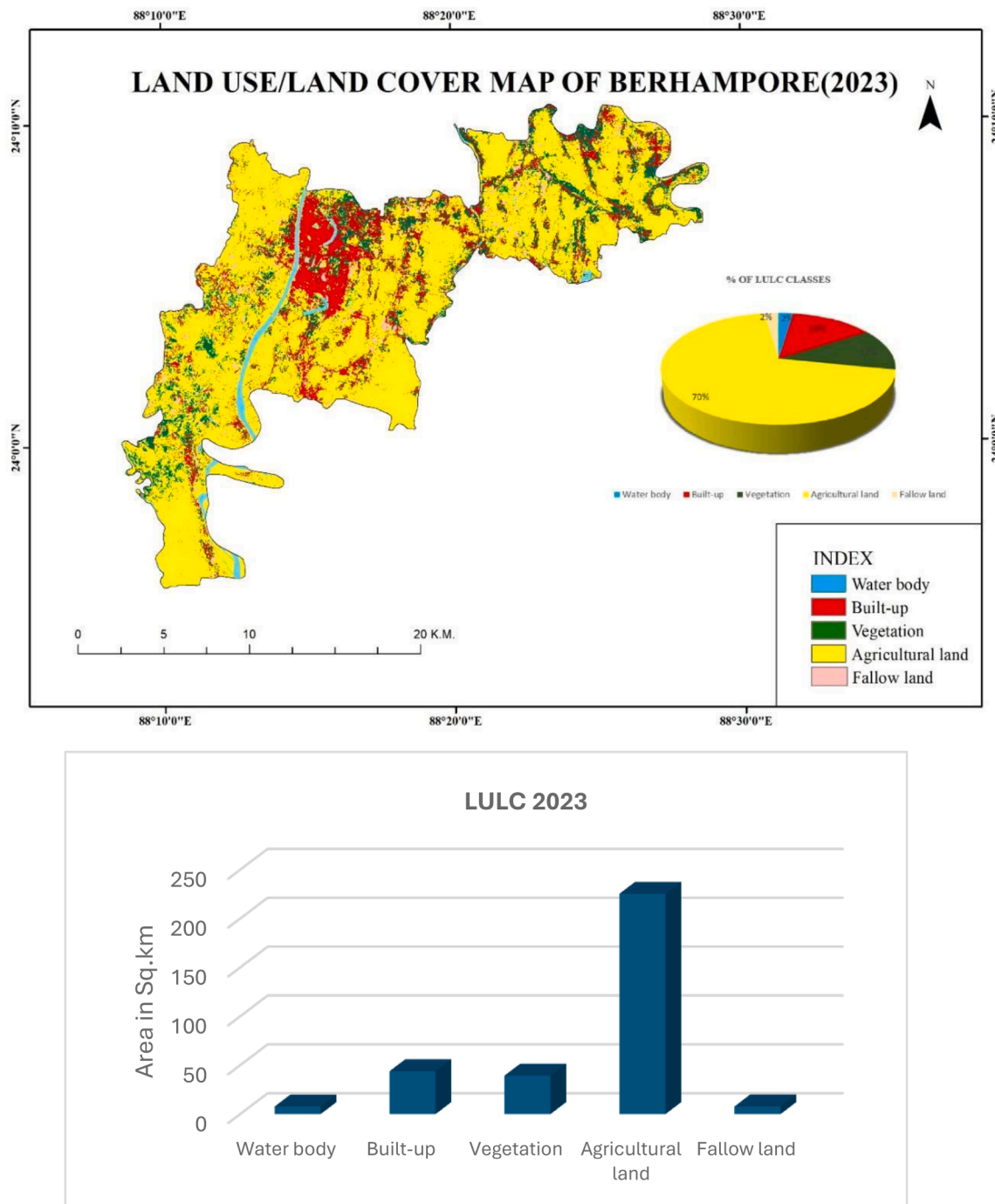


Fig. 6. LULC of the study area in 2023 (a) Map; (b) Area in sq. km.

spatially explicit model in the analysis and prediction of the LULC dynamics in medium-sized cities. The conceptual extension that has been proposed based on hybrid deep learning architectures (CNN-LSTM) is a roadmap to further enhancing the spatio-temporal prediction in future research.

The results highlight the need to integrate predictive LULC modelling in urban planning to promote proactive and sustainable land management. To enhance the real-time monitoring and climate-resilient urban planning, future studies ought to combine socio-economic motivators, high-frequency satellite images, and practical deep learning systems.

CRedit authorship contribution statement

Debabrata Nandi: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Rakesh Ranjan Thakur:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Dillip Kumar Bera:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Alok Kumar Pati:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Priyanka Mishra:** Writing – review & editing, Supervision. **Fahad Alshehri:** Writing – review & editing. **Mohamed**

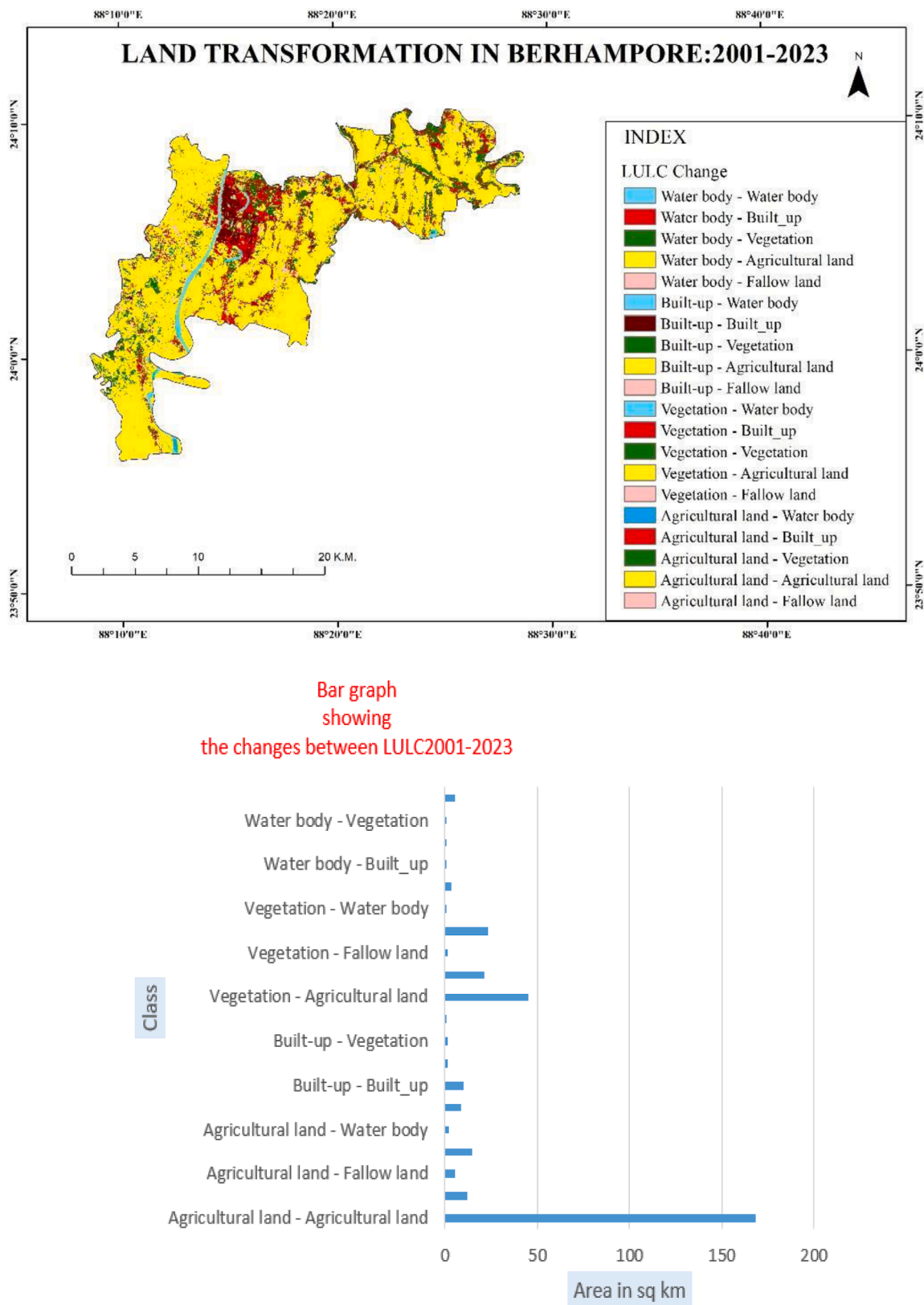


Fig. 7. LULC changes between 2001 and 2023 (a) Map; (b) Changes in area (sq. km.).

Zhran: Writing – review & editing, Supervision, Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

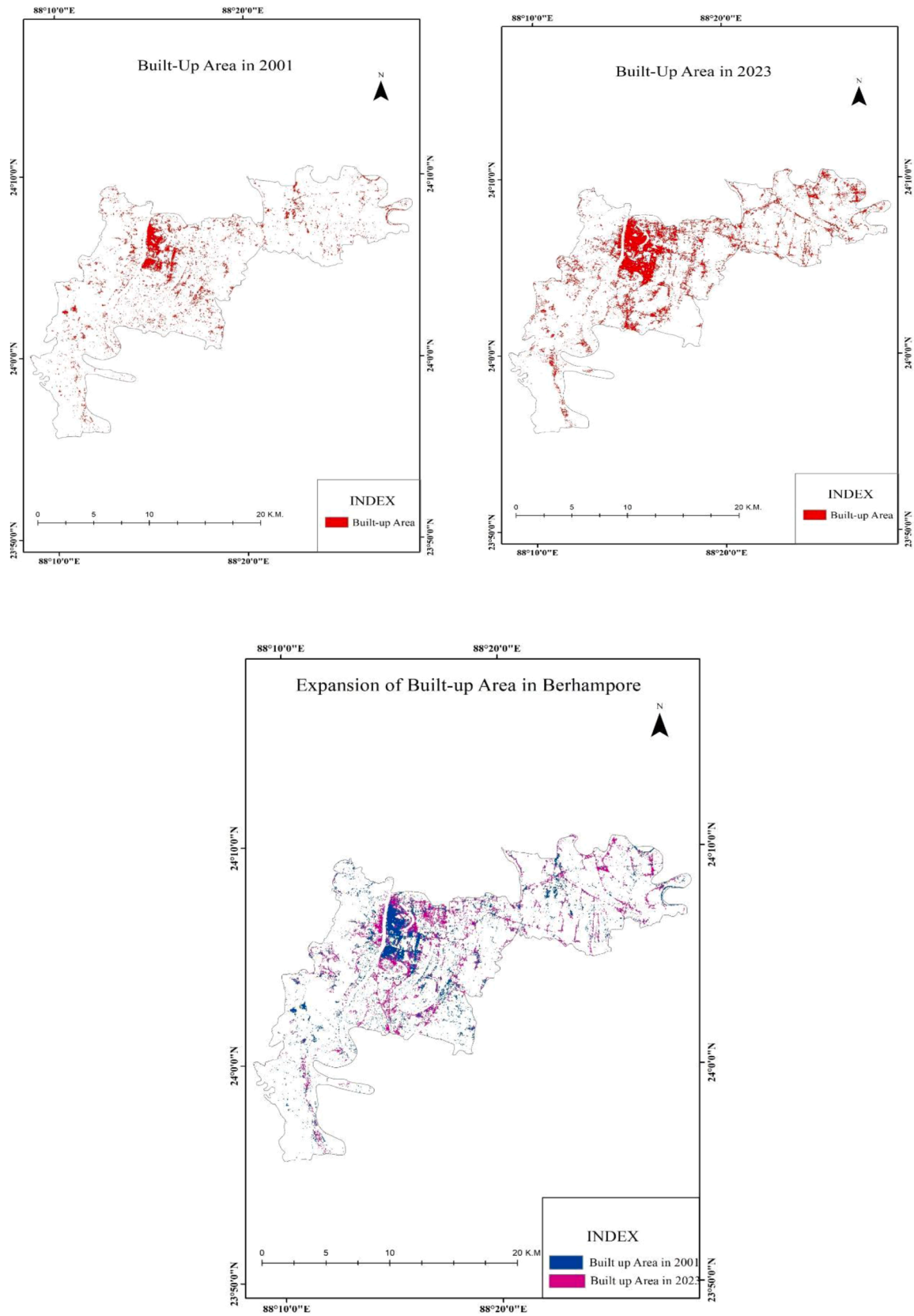


Fig. 8. Built-Up Area (a) 2001; (b) 2023; (c) Expansion trend.

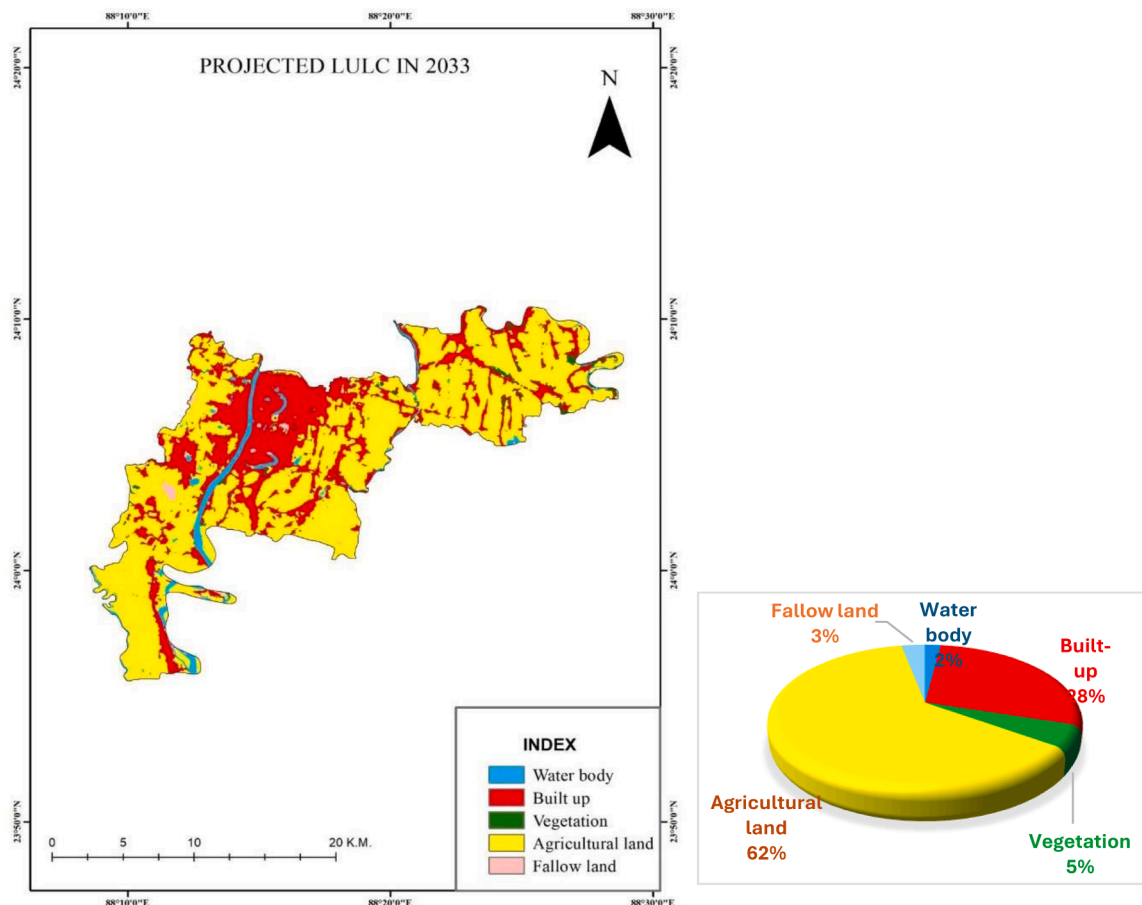


Fig. 9. Forecast of LULC in 2033.

the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] A. Tariq, H. Shu, S. Siddiqui, M. Imran, M. Farhan, Monitoring land use and land cover changes using geospatial techniques: a case study of Fateh Jang, Attock, Pakistan, *Geogr. Environ. Sustain.* 14 (1) (2021) 41–52, <https://doi.org/10.24057/2071-9388-2020-117>.
- [2] E.T. Iortyom, J.T. Semaka, J.I. Abawua, Spatial expansion of urban activities and agricultural lands encroachment in Makurdi metropolis, *Eur. J. Environ. Earth Sci.* 1 (6) (2020), <https://doi.org/10.24018/ejgeo.2020.1.6.89>.
- [3] M.B. Eppinga, H.J. de Boer, M.O. Reader, J.M. Anderies, M.J. Santos, Environmental change and ecosystem functioning drive transitions in social-ecological systems: a stylized modelling approach, *Ecol. Econ.* 211 (2023) 107861, <https://doi.org/10.1016/j.ecolecon.2023.107861>.
- [4] M. Scheffer, S.R. Carpenter, V. Dakos, E.H. van Nes, Generic indicators of ecological resilience: inferring the chance of a critical transition, *Annu. Rev. Ecol. Syst.* 46 (1) (2015) 145–167, <https://doi.org/10.1146/annurev-ecolsys-112414-054242>.
- [5] P.H. Verburg, O. Mertz, K. Erb, H. Haberl, W. Wu, Land system change and food security: towards multi-scale land system solutions, *Curr. Opin. Environ. Sustain.* 5 (5) (2013) 494–502, <https://doi.org/10.1016/j.cosust.2013.07.003>.
- [6] S. Krätli, C. Huelsebusch, S. Brooks, B. Kaufmann, Pastoralism: a critical asset for food security under global climate change, *Anim. Front.* 3 (1) (2013) 42–50, <https://doi.org/10.2527/af.2013-0007>.
- [7] J. Mandal, N. Ghosh, A. Mukhopadhyay, Urban growth dynamics and changing land-use land-cover of megacity Kolkata and its environs, *J. Indian Soc. Remote Sens.* 47 (10) (2019) 1707–1725, <https://doi.org/10.1007/s12524-019-01020-7>.
- [8] N. Khan, R.L. Ray, G.R. Sargani, M. Ihtisham, M. Khayyam, S. Ismail, Current progress and future prospects of agriculture technology: gateway to sustainable agriculture, *Sustainability* 13 (9) (2021) 4883, <https://doi.org/10.3390/su13094883>.
- [9] M.D. McGinnis, E. Ostrom, Social-ecological system framework: initial changes and continuing challenges, *Ecol. Soc.* 19 (2) (2014), <https://doi.org/10.5751/ES-06387-190230>.
- [10] M.K. Leta, T.A. Demissie, J. Tränckner, Modeling and prediction of land use land cover change dynamics based on land change modeler (LCM) in Nashe watershed, Upper Blue Nile Basin, Ethiopia, *Sustain.* 13 (7) (2021) 3740, <https://doi.org/10.3390/su13073740>.
- [11] S. Gaur, R. Singh, A comprehensive review on land use/land cover (LULC) change modeling for urban development: current status and future prospects, *Sustainability* 15 (2) (2023) 903, <https://doi.org/10.3390/su15020903>.
- [12] W. Khalid, S.K. Shamim, A. Ahmad, Synergistic approach for land use and land cover dynamics prediction in Uttarakhand using cellular automata and artificial neural network, *Geomatica* 76 (2) (2024) 100017, <https://doi.org/10.1016/j.geomat.2024.100017>.
- [13] M. Tanku, B. Woldetensae, Industrial development and urban spatial planning practices: the case of Galan and Dukem cities in Ethiopia, *Heliyon* 9 (7) (2023), <https://doi.org/10.1016/j.heliyon.2023.e17554>.
- [14] D. Kalfas, S. Kalogiannidis, F. Chatzitheodoridis, E. Toska, Urbanisation and land use planning for achieving the sustainable development goals (SDGs): a case study of Greece, *Urban. Sci.* 7 (2) (2023) 43, <https://doi.org/10.3390/urbansci7020043>.
- [15] M. Pandey, R.R. Thakur, D. Nandi, D.K. Bera, R. Beuria, M. Kumari, A. M. Kasawnea, M. Zhran, Geospatial monitoring of environmental sustainability: a remote sensing-based approach for assessing mining-induced impacts in eastern India, *Results. Eng.* 26 (2025) 104692, <https://doi.org/10.1016/j.rineng.2025.104692>.
- [16] S. Das, D. Nandi, R.R. Thakur, D.K. Bera, D. Behera, B. Durin, V. Cetl, A novel approach for ex situ water quality monitoring using the Google Earth Engine and spectral indices in Chilika Lake, Odisha, India, *ISPRS Int. J. Geo-Inf.* 13 (11) (2024) 381, <https://doi.org/10.3390/ijgi13110381>.
- [17] D. Nandi, R.R. Das, D. Singh, I. Bera, P.S. Mishra, P.C. Sahu, K.L. Mohanta, A comprehensive investigation on the effect of UHI in Baripada city using

- geospatial technique and MLTHP model, *Environ. Qual. Manag.* 34 (1) (2024), <https://doi.org/10.1002/tqem.22233>.
- [18] A. Alzahrani, N. Aldossary, J. Alghamdi, Observing the dynamics of urban growth of Al-Baha city using GIS (2006–2021), *Alexandria Eng. J.* 95 (2024) 114–131, <https://doi.org/10.1016/j.aej.2024.03.096>.
 - [19] D.G. Costa, J.C.N. Bittencourt, F. Oliveira, J.P.J. Peixoto, T.C. Jesus, Achieving sustainable smart cities through geospatial data-driven approaches, *Sustainability* 16 (2) (2024) 640, <https://doi.org/10.3390/su16020640>.
 - [20] V. Nasiri, A. Deljouei, F. Moradi, S.M.M. Sadeghi, S.A. Borz, Land use and land cover mapping using Sentinel-2, Landsat-8 satellite images, and Google Earth Engine: a comparison of two composition methods, *Remote Sens.* 14 (9) (2022), <https://doi.org/10.3390/rs14091977>, 1977.
 - [21] C. Zhang, X. Li, Land use and land cover mapping in the era of big data, *Land. (Basel)* 11 (10) (2022) 1692, <https://doi.org/10.3390/land11101692>.
 - [22] A.A. Saim, M. Aly, Enhancing tree species mapping in Arkansas forests through machine learning and satellite data fusion: a Google Earth Engine-based approach, *J. Geovisualization Spat. Anal.* 9 (1) (2025), <https://doi.org/10.1007/s41651-025-00020-9>.
 - [23] D. Yu, C. Fang, Urban remote sensing with spatial big data: a review and renewed perspective of urban studies in recent decades, *Remote Sens.* 15 (5) (2023) 1307, <https://doi.org/10.3390/rs15051307>.
 - [24] K. Hussain, K. Mehmood, S. Yujun, T. Badshah, S.A. Anees, F. Shahzad, N. Nooruddin, J. Ali, M. Bilal, Analysing LULC transformations using remote sensing data: insights from a multilayer perceptron neural network approach, *Ann. GIS.* (2024) 1–28, <https://doi.org/10.1080/19475683.2024.2343399>.
 - [25] Z. Khan, W. Khalid, S.A. Ali, S.K. Shamim, A. Ahmad, Analysing the potential impacts of land use land cover (LULC) transformation on present and future carbon sequestration capabilities in the central Himalayas, *Discov. Geosci.* 2 (1) (2024), <https://doi.org/10.1007/s44288-024-00097-z>.
 - [26] M.M.H. Seyam, M.R. Haque, M.M. Rahman, Identifying the land use land cover (LULC) changes using remote sensing and GIS approach: a case study at Bhaluka in Mymensingh, Bangladesh, *Case Stud. Chem. Environ. Eng.* 7 (2023) 100293, <https://doi.org/10.1016/j.csee.2022.100293>.
 - [27] Z. Li, B. Chen, S. Wu, M. Su, J.M. Chen, B. Xu, Deep learning for urban land use category classification: a review and experimental assessment, *Remote Sens. Environ.* 311 (2024) 114290, <https://doi.org/10.1016/j.rse.2024.114290>.
 - [28] S.A.H. Selmy, D.E. Kucher, G. Mozgeris, A.R.A. Moursy, R. Jimenez-Ballesta, O. D. Kucher, M.E. Fadl, A.A. Mustafa, Detecting, analyzing, and predicting land use/land cover (LULC) changes in arid regions using Landsat images, CA–Markov hybrid model, and GIS techniques, *Remote Sens.* 15 (23) (2023) 5522, <https://doi.org/10.3390/rs15235522>.
 - [29] Z. Hassan, R. Shabbir, S.S. Ahmad, A.H. Malik, N. Aziz, A. Butt, S. Erum, Dynamics of land use and land cover change (LULCC) using geospatial techniques: a case study of Islamabad, Pakistan, *SpringerPlus* 5 (1) (2016) 812, <https://doi.org/10.1186/s40064-016-2414-z>.
 - [30] A.K. Pati, A.R. Tripathy, D. Nandi, R.R. Thakur, B. Durin, D. Dogančić, O. Fetoshi, Fluoride risk prognostication: a pioneering ensemble machine learning approach for groundwater contamination prediction in parts of the east coast of India, *Water* 17 (6) (2025) 909, <https://doi.org/10.3390/w17060909>.
 - [31] M. Kasahun, A. Legesse, Machine learning for urban land use/cover mapping: comparison of artificial neural network, random forest and support vector machine, a case study of Dilla Town, Heliyon. 10 (20) (2024), <https://doi.org/10.1016/j.heliyon.2024.e39146> e39146.
 - [32] M. Asif, J.H. Kazmi, A. Tariq, N. Zhao, R. Guluzade, W. Soufan, K.F. Almutairi, A. E. Sabagh, M. Aslam, Modelling of land use and land cover changes and prediction using CA–Markov and random forest, *Geocarto Int.* 38 (1) (2023), <https://doi.org/10.1080/10106049.2023.2210532>.
 - [33] S.W. Wang, L. Munkhnasan, W. Lee, Land use and land cover change detection and prediction in Bhutan's high altitude city of Thimphu using cellular automata and Markov chain, *Environ. Chall.* 2 (2021) 100017, <https://doi.org/10.1016/j.envc.2020.100017>.
 - [34] A. Azabdaftari, F. Sunar, Predicting urban tomorrow: CA–Markov modeling and district evolution, *Earth. Sci. Inform.* (2024), <https://doi.org/10.1007/s12145-024-01340-4>.
 - [35] K. de Oliveira Barros, C.A.A.S. Ribeiro, G.E. Marcatti, A.S. Lorenzon, N.L.M. de Castro, G.F. Domingues, J.R. de Carvalho, A.R.D. Santos, Markov chains and cellular automata to predict environments subject to desertification, *J. Environ. Manag.* 225 (2018) 160–167, <https://doi.org/10.1016/j.jenvman.2018.07.064>.
 - [36] W. Khalid, S.K. Shamim, A. Ahmad, Synergistic approach for land use and land cover dynamics prediction in Uttarakhand using cellular automata and artificial neural network, *Geomatica* 76 (2) (2024) 100017, <https://doi.org/10.1016/j.geomat.2024.100017>.
 - [37] D. Gounaridis, I. Chorianopoulos, E. Symeonakis, S. Koukoulas, A random forest–cellular automata modelling approach to explore future land use/cover change in Attica (Greece), under different socio-economic realities and scales, *Sci. Total. Environ.* 646 (2019) 320–335, <https://doi.org/10.1016/j.scitotenv.2018.07.302>.
 - [38] Q. Xu, A. Zhu, J. Liu, Land-use change modeling with cellular automata using land natural evolution unit, *Catena* 224 (2023) 106998, <https://doi.org/10.1016/j.catena.2023.106998>.
 - [39] A. Gharaibeh, A. Shaamala, R. Obeidat, S. Al-Kofahi, Improving land-use change modeling by integrating ANN with cellular automata–Markov chain model, *Heliyon.* 6 (9) (2020) e05092, <https://doi.org/10.1016/j.heliyon.2020.e05092>.
 - [40] A.B. Rashid, A.K. Kausik, A. Khandoker, S.N. Siddique, Integration of artificial intelligence and IoT with UAVs for precision agriculture, *Hybrid Adv.* (2025) 100458, <https://doi.org/10.1016/j.hybadv.2025.100458>.
 - [41] N. Haripavan, S. Dey, C.H.M. Chandana, Integration of geospatial techniques and machine learning in land parcel prediction, *Geosyst. Geoenviron.* 4 (2) (2025) 100371, <https://doi.org/10.1016/j.geogeo.2025.100371>.
 - [42] S.S. Mukonza, J.-L. Chiang, Meta-analysis of satellite observations for United Nations Sustainable Development Goals: exploring the potential of machine learning for water quality monitoring, *Environments* 10 (10) (2023) 170, <https://doi.org/10.3390/environments10100170>.
 - [43] F. Deng, J. Huang, X. Yuan, C. Cheng, L. Zhang, Performance and efficiency of machine learning algorithms for analyzing rectangular biomedical data, *Lab. Invest.* 101 (4) (2021) 430–441, <https://doi.org/10.1038/s41374-020-00525-x>.
 - [44] I.H. Sarker, Machine learning: algorithms, real-world applications and research directions, *SN. Comput. Sci.* 2 (3) (2021) 160, <https://doi.org/10.1007/s42979-021-00592-x>.
 - [45] T. Adugna, W. Xu, J. Fan, Comparison of random forest and support vector machine classifiers for regional land cover mapping using coarse resolution FY-3C images, *Remote Sens.* 14 (3) (2022) 574, <https://doi.org/10.3390/rs14030574>.
 - [46] E. Raczko, B. Zagajewski, Comparison of support vector machine, random forest and neural network classifiers for tree species classification on airborne hyperspectral APEX images, *Eur. J. Remote Sens.* 50 (1) (2017) 144–154, <https://doi.org/10.1080/22797254.2017.1299557>.
 - [47] B. Aftab, Z. Wang, S. Wang, Z. Feng, Application of a multi-layer perceptron and Markov chain analysis-based hybrid approach for predicting and monitoring LULCC patterns using random forest classification in Jhelum District, Punjab, Pakistan, *Sensors* 24 (17) (2024) 5648, <https://doi.org/10.3390/s24175648>.
 - [48] M. Jain, Future land use and land cover simulations with cellular automata-based artificial neural network: a case study over Delhi megacity (India), *Heliyon.* 10 (14) (2024) e34662, <https://doi.org/10.1016/j.heliyon.2024.e34662>.
 - [49] M.A. Siddique, Y. Wang, N. Xu, N. Ullah, P. Zeng, The spatiotemporal implications of urbanisation for urban heat islands in Beijing: a predictive approach based on CA–Markov modeling (2004–2050), *Remote Sens.* 13 (22) (2021) 4697, <https://doi.org/10.3390/rs13224697>.
 - [50] L. Alzubaidi, J. Zhang, A.J. Humaidi, A. Al-Dujaili, Y. Duan, O. Al-Shamma, J. Santamaría, M.A. Fadhel, M. Al-Amidie, L. Farhan, Review of deep learning: concepts, CNN architectures, challenges, applications, future directions, *J. Big. Data* 8 (1) (2021) 53, <https://doi.org/10.1186/s40537-021-00444-8>.
 - [51] T.O. Ogunbode, V.O. Oyebamiji, D.O. Sanni, E.O. Akinwale, F.O. Akinluyi, Environmental impacts of urban growth and land use changes in tropical cities, *Front. Sustain. Cities.* 6 (2025), <https://doi.org/10.3389/frsc.2024.1481932>.
 - [52] C.T. Lloyd, A. Sorichetta, A.J. Tatem, High resolution global gridded data for use in population studies, *Sci. Data* 4 (1) (2017), <https://doi.org/10.1038/sdata.2017.1>.
 - [53] A.-A. Kafy, M.N.H. Naim, G. Subramanyam, A. Faisal, N.U. Ahmed, A.A. Rakib, M. A. Kona, G.S. Sattar, Cellular automata approach in dynamic modelling of land cover changes using RapidEye images in Dhaka, Bangladesh, *Environ. Chall.* 4 (2021) 100084, <https://doi.org/10.1016/j.envc.2021.100084>.
 - [54] A.E. Raftery, H. Ševčíková, Probabilistic population forecasting: short to very long-term, *Int. J. Forecast.* 39 (1) (2021) 73–97, <https://doi.org/10.1016/j.ijforecast.2021.09.001>.
 - [55] R. Muhammad, W. Zhang, Z. Abbas, F. Guo, L. Gwiazdzinski, Spatiotemporal change analysis and prediction of future land use and land cover changes using QGIS MOLUSCE plugin and remote sensing big data: a case study of Linyi, China, *Land* 11 (3) (2022) 419, <https://doi.org/10.3390/land11030419>.
 - [56] A. Zhang, A. Tariq, A. Qudus, I. Naz, R.W. Aslam, E. Barboza, S. Ullah, M. Abdullah-Al-Wadud, Spatio-temporal analysis of urban expansion and land use dynamics using Google Earth Engine and predictive models, *Sci. Rep.* 15 (1) (2025), <https://doi.org/10.1038/s41598-025-92034-4>.
 - [57] R.N. Masolele, V. De Sy, M. Herold, D. Marcos, J. Verbesselt, F. Gieseke, A. G. Mullissa, C. Martius, Spatial and temporal deep learning methods for deriving land-use following deforestation: a pan-tropical case study using Landsat time series, *Remote Sens. Environ.* 264 (2021) 112600, <https://doi.org/10.1016/j.rse.2021.112600>.
 - [58] R. Wahdatyar, M.F. Khokhar, S. Ahmad, M.U. Rahil, M.A. Stanikzai, J.A. Khan, N. Kamran, Exploring the dynamics and future projections of land use land cover changes by exploiting geospatial techniques: a case study of the Kabul River Basin, *Heliyon.* 10 (20) (2024), <https://doi.org/10.1016/j.heliyon.2024.e39020> e39020.
 - [59] O. Bhungeni, A. Ramjatan, M. Gebreslasie, Evaluating machine-learning algorithms for mapping LULC of the uMngeni catchment area, in: KwaZulu-Natal, *Remote Sens.*, 16, 2024, p. 2219, <https://doi.org/10.3390/rs16122219>.
 - [60] X. He, Q. Li, R. Wang, K. Chen, A spatio-temporal feature trajectory clustering algorithm based on deep learning, *Electronics* 11 (15) (2022) 2283, <https://doi.org/10.3390/electronics11152283>.
 - [61] X. Zhao, P. Wang, S. Gao, M. Yasir, Q.U. Islam, Combining LSTM and PLUS models to predict future urban land use and land cover change: a case in Dongying City, China, *Remote Sens.* 15 (9) (2023) 2370, <https://doi.org/10.3390/rs15092370>.
 - [62] B. Sajjan, V.N. Mishra, S. Kanga, G. Meraj, S.K. Singh, P. Kumar, Cellular automata-based artificial neural network model for assessing past, present, and future land use/land cover dynamics, *Agronomy* 12 (11) (2022) 2772, <https://doi.org/10.3390/agronomy12112772>.
 - [63] P. Lukas, A.M. Melesse, T.T. Kenea, Prediction of future land use/land cover changes using a coupled CA–ANN model in the Upper Omo–Gibe River Basin, Ethiopia, *Remote Sens.* 15 (4) (2023) 1148, <https://doi.org/10.3390/rs15041148>.
 - [64] A. Choubineh, J. Chen, F. Coenen, F. Ma, Applying Monte Carlo dropout to quantify the uncertainty of skip connection-based convolutional neural networks optimized by big data, *Electronics* 12 (6) (2023) 1453, <https://doi.org/10.3390/electronics12061453>.
 - [65] B. Feizizadeh, K.M. Alajujeh, T. Lakes, T. Blaschke, D. Omarzadeh, A comparison of the integrated fuzzy object-based deep learning approach and three machine

- learning techniques for land use/cover change monitoring and environmental impacts assessment, *Glsci. Remote Sens.* 58 (8) (2021) 1543–1570, <https://doi.org/10.1080/15481603.2021.2000350>.
- [66] M.N. Rahman, M.M. Saleheen, A.R.M.T. Islam, M.S. Sohel, M.S. Islam, Assessing and predicting the dynamics of land use/land cover in northern Bangladesh using cellular automata–Markov chain model, *Geol. Ecol. Landsc.* (2024) 1–17, <https://doi.org/10.1080/24749508.2024.2429218>.
- [67] P.A. Palanisamy, J. Zawadzka, K. Jain, S. Bonafoni, A. Tiwari, Assessing diurnal land surface temperature variations across landcover and local climate zones: implications for urban planning and mitigation strategies on socio-economic factors, *Sustain. Cities Soc.* 116 (2024) 105880, <https://doi.org/10.1016/j.scs.2024.105880>.
- [68] M. Panigrahi, A. Sharma, Urban growth dynamics and its influence on land surface temperature in Bhubaneswar metropolitan city: a 1990–2021 analysis, *Discov. Appl. Sci.* 7 (2) (2025) 118, <https://doi.org/10.1007/s42452-025-06535-y>.
- [69] X. Cheng, Y. Sun, W. Zhang, Y. Wang, X. Cao, Y. Wang, Application of deep learning in multitemporal remote sensing image classification, *Remote Sens.* 15 (15) (2023) 3859, <https://doi.org/10.3390/rs15153859>.
- [70] L. Ma, Y. Liu, X. Zhang, Y. Ye, G. Yin, B.A. Johnson, Deep learning in remote sensing applications: a meta-analysis and review, *ISPRS. J. Photogramm. Remote Sens.* 152 (2019) 166–177, <https://doi.org/10.1016/j.isprsjprs.2019.04.015>.
- [71] A. Mohammed, R. Kora, An effective ensemble deep learning framework for text classification, *J. King Saud Univ. Comput. Inf. Sci.* 34 (10) (2021) 8825–8837, <https://doi.org/10.1016/j.jksuci.2021.11.001>.
- [72] G. Yan, C. Yu, Y. Bai, A new hybrid ensemble deep learning model for train axle temperature short-term forecasting, *Machines* 9 (12) (2021) 312, <https://doi.org/10.3390/machines9120312>.
- [73] E.A. Abdelsamie, A.A. Mustafa, A.S. El-Sorogy, H.F. Maswada, S.A. Almadani, M. S. Shokr, A.I. El-Desoky, J.E.M. De Larriva, Current and potential land use/land cover (LULC) scenarios in dry lands using a CA–Markov simulation model and the classification and regression tree (CART) method: a cloud-based Google Earth Engine (GEE) approach, *Sustainability*. 16 (24) (2024) 11130, <https://doi.org/10.3390/su162411130>.
- [74] S. Ghosh, D. Kumar, R. Kumari, Cloud-based large-scale data retrieval, mapping, and analysis for land monitoring applications with Google Earth Engine (GEE), *Environ. Chall.* 9 (2022) 100605, <https://doi.org/10.1016/j.envc.2022.100605>.
- [75] M.W.N. Shahfahad, T. Das, S. Talukdar, M.S. Asgher, N. Asif, A. Rahman, Prediction of land use changes at a metropolitan city using integrated cellular automata: past and future, *Geol. Ecol. Landsc.* 8 (3) (2022) 287–305, <https://doi.org/10.1080/24749508.2022.2132010>.
- [76] A.E. Al-Dousari, A. Mishra, S. Singh, Land use land cover change detection and urban sprawl prediction for Kuwait metropolitan region using multi-layer perceptron neural networks (MLPNN), *Egypt. J. Remote Sens. Space Sci.* 26 (2) (2023) 381–392, <https://doi.org/10.1016/j.ejrs.2023.05.003>.
- [77] M.F. Baig, M.R.U. Mustafa, I. Baig, H.B. Takaijudin, M.T. Zeshan, Assessment of land use land cover changes and future predictions using CA–ANN simulation for Selangor, Malaysia, *Water* 14 (3) (2022) 402, <https://doi.org/10.3390/w14030402>.
- [78] E. Rustiadi, A.E. Pravitasari, Y. Setiawan, S.P. Mulya, D.O. Pribadi, N. Tsutsumida, Impact of continuous Jakarta megacity urban expansion on the formation of the Jakarta–Bandung conurbation over the rice farm regions, *Cities*. 111 (2020) 103000, <https://doi.org/10.1016/j.cities.2020.103000>.
- [79] B. Surya, D.N.A. Ahmad, H.H. Sakti, H. Sahban, Land use change, spatial interaction, and sustainable development in the metropolitan urban areas, South Sulawesi Province, Indonesia, *Land* 9 (3) (2020) 95, <https://doi.org/10.3390/land9030095>.